

Plant-based foods as a source of lipotropes for human nutrition: a survey of in vivo studies.

Increased consumption of plant products is associated with lower chronic disease prevalence. This is attributed to the great diversity of healthy phytochemicals present in these foods. The most investigated physiological effects have been their antioxidant, anti-carcinogenic, hypolipidemic, and hypoglycemic properties. Although less studied in humans, some compounds were very early on shown to be lipotropic in animals, i.e., the capacity to hasten the removal of fat from liver and/or reduce hepatic lipid synthesis or deposits by mainly increasing phospholipid synthesis via the transmethylation pathway for triglyceride-rich lipoprotein exportation from the liver and enhanced fatty acid β -oxidation and/or down- and up-regulation of genes involved in lipogenic and fatty acid oxidation enzyme synthesis, respectively. The main plant lipotropes are choline, betaine, myo-inositol, methionine, and carnitine. Magnesium, niacin, pantothenate, and folates also indirectly support the overall lipotropic effect. The exhaustive review of rat studies investigating phytochemical effect on hepatic lipid metabolism suggests that some fatty acids, acetic acid, melatonin, phytic acid, some fiber compounds, oligofructose, resistant starch, some phenolic acids, flavonoids, lignans, stilbenes, curcumin, saponins, coumarin, some plant extracts, and some solid foods may be lipotropic. However, this remains to be confirmed in humans, for whom intervention studies are practically non-existent. Supplemental materials are available for this article. Go to the publisher's online edition of *Critical Reviews in Food Science and Nutrition*® to view the free supplemental file.